Dr. Rasdeep Kour

Assistant Professor, Department of Sciences (Botany),
Khalsa College for Women, Amritsar, India–143005
+91 8130296804   kourrasdeep@gmail.com

Education

Guru Nanak Dev University, Amritsar, India
Ph.D. Botany Degree Awarded: 13/12/2024
04/09/2018 – 13/12/2024

Guru Nanak Dev University, Amritsar, India
M.Sc. Botany Degree Awarded: 15/06/2017
01/07/2015 – 15/06/2017

Govt. Degree College, Poonch, Jammu and Kashmir, India
B.Sc. Degree Awarded: 09/06/2015
01/07/2012 – 09/06/2015

Research Experience

Guru Nanak Dev University, Amritsar, India
Ph.D. Botany 04/09/2018 – 13/12/2024

Research Objective: To evaluate the antioxidant, genoprotective and anticancer potential of *Cassia fistula L.* bark phytoconstituents using in vitro and in vivo models.

The researcher has worked on the extraction and isolation of *C. fistula* bark phytoconstituents through bioactivity-guided fractionation and structural characterization. The study focused on assessing antioxidant, genoprotective, and selective cytotoxic effects, elucidating apoptosis-mediated pathways, and developing a liposomal formulation to enhance bioavailability and in vivo tumor suppression, aiming to translate phytochemicals into safe, mechanism-driven chemopreventive and chemotherapeutic candidates.

Doctoral Dissertation Title: Assessment of Antioxidant, Genoprotective and Anticancer potential of *Cassia fistula L.* bark

Peer-Reviewed Publications

1. Rasdeep Kour, Neha Sharma, Sheikh Showkat, Sunil Sharma, Kommu Nagaiah, Subodh Kumar and Satwinderjeet Kaur*. Methanolic fraction of *Cassia fistula L.* bark exhibits potential to combat oxidative stress and possess antiproliferative activity. *Journal of Toxicology and Environmental Health, Part A*, 2023, 86(9), 296–312.

2. Rasdeep Kour, Neha Sharma, Mangaljeet Singh, Subodh Kumar, and Satwinderjeet Kaur*. *Cassia fistula L.* bark fraction modulated GSK3β/p53 expression for mitochondrial mediated apoptosis in HeLa cells. *South African Journal of Botany*, 2024, 168, 46–60.

3. Simrandeep Kaur, Rasdeep Kour, Majed Alrobaian, Ashish Kumar, Aman Agrawal, Sandeep Kaur, Vijai Malik, Satwinderjeet Kaur and Sheikh Showkat Ahmad. Phytochemical profiling, antioxidant, DNA protective and antiproliferative potential of methanolic fraction of Hemidesmus indicus (L.) R. Br. roots: in vitro and in silico studies. *In vitro models*, 2026.

4. Neha Sharma, Sandeep Kaur, Rasdeep Kour, Vaseem Raja, Satwinderjeet Kaur, Showket Ahmad and Khalid Rehman Hakeem*. Chemopreventive potential of quinones: Molecular targets and therapeutic potential. *Mol Biol Rep*, 2026, 53, 840.

5. Sukhvinder Dhiman, Gulshan Kumar, Rasdeep Kour, Satwinderjeet Kaur, Vijay Luxami, Prabhpreet Singh and Subodh Kumar*. A red-emissive tripodal nanoprobe for the discrimination of serum albumin with conformational change from Y- to ω-like and probing mitochondrial viscosity. *Journal of Material Chemistry B*, 2025, 13, 5634–5642.

6. Omkar Bains, Ashish Kumar, Raj Kamal*, Rasdeep Kour, Simrandeep Kaur, Satwinderjeet Kaur, Raman Jangra, Purshotam Sharma, Ravinder Kumar. Synthesis, anticancer evaluation, in-silico ADMET and molecular docking studies for tailored pyrazolo-benzothiazole hybrids. *Asian Journal of Organic Chemistry*, 2024, e202400187.

7. Sheikh Showkat Ahmad, Chandni Garg, Rasdeep Kour, Aashaq Hussain Bhat, Vaseem Raja, Sumit G. Gandhi, Farid S. Ataya, Dalia Fouad, Arunkumar Radhakrishnan and Satwinderjeet Kaur*. Metabolomic insights and bioactive efficacies of *Tragopogon dubius* root fractions: Antioxidant and antiproliferative assessments. *Heliyon*, 2024, 10(16), e34746.

8. Neha Sharma, Rasdeep Kour, Shagun Verma, Vandana Sharma, Deepika Singh, Sumit G. Gandhi, Vaseem Raja*, Satwinderjeet Kaur*, Naveen Kumar, Khalid Mashay Al-Anazi, Mohammad Abul Farah. Growth inhibitory potential of quinone-rich fraction of *Plumbago zeylanica L.* against MG-63 osteosarcoma cells via Bax/Bcl-2 modulation. *Journal of King Saud University-Science*, 2024, 36(9), 103405.

9. Navdeep Kaur, Rasdeep Kour, Sahil Gasso, Satwinder Singh Marok, Satwinderjeet Kaur, Aman Mahajan and Prabhpreet Singh*. Near-IR intracellular ratiometric 'turn-on' discrimination of H_2S/Cys and low-cost test kits for ppm level detection of H_2S gas. *Journal of Photochemistry and Photobiology A: Chemistry*, 2023, 435, 114345.

10. Navdeep Kaur, Rasdeep Kour, Satwinderjeet Kaur and Prabhpreet Singh* (2023). Perylene diimide-based sensors for multiple analyte sensing (Fe^{2+}/H_2S/dopamine and Hg^{2+}/Fe^{2+}): cell imaging and INH, XOR, and encoder logic. *Analytical Methods*, 2023, 15(19), 2391–2398.

11. Rajdeep Kaur, Rasdeep Kour, Satwinderjeet Kaur, Prabhpreet Singh*. An activatable AIEgen for imaging of endogenous H_2S: Reticulation based H_2S gas sensor. *Journal of Photochemistry and Photobiology A: Chemistry*, 2023, 444, 114995.

12. Navdeep Kaur, Rasdeep Kour, Satwinderjeet Kaur and Prabhpreet Singh*. Intelligent Colorimetric Sensor for Discriminative Detection of Pd^{2+}/Pd^0 and PPh_3 at Different Wavelength Readouts. *ChemistrySelect*, 2023, 28(26), e202300693.

13. Falak Thakral, Ajay Kumar, Manik Sharma, Rasdeep Kour, Satwinderjeet Kaur, Moyad Shahwan, Ritu Chauhan, Gurpreet Kaur Bhatia, Hardeep Singh Tuli*. Green Synthesized Zinc Oxide Nanoparticles induced Apoptosis in Human Cervical (HeLa) and Breast (MCF-7) Cancer Cell Lines. *Asian Journal of Chemistry*, 2023, 35(8), 1979–1984.

14. Navdeep Kaur, Rasdeep Kour, Satwinderjeet Kaur and Prabhpreet Singh*. Near-IR intracellular Pd^0/Pd^{2+} triggered de-allylation switch based on self-assembled perylene diimide. *Journal of Photochemistry and Photobiology A: Chemistry*, 2023, 445, 115068.

15. Neha Sharma, Shubham Thakur, Rasdeep Kour, Deepika, Ajay Kumar, Ajaz Ahmad, Prashant Kaushik, Vaseem Raja*, Subheet Kumar Jain and Satwinderjeet Kaur*. *Plumbago zeylanica L.* exhibited potent anticancer activity in Ehrlich ascites carcinoma bearing Swiss albino mice. *Journal of King Saud University-Science*, 2023, 35(9), 102932.

16. Manjot Kaur, Riya Shivgotra, Shubham Thakur, Rasdeep Kour, Manpreet Singh, Navid Reza Shahtaghi, Satwinderjeet Kaur and Subheet Kumar Jain*. A novel nanoemulgel formulation of Luliconazole with augmented antifungal efficacy: In-vitro, in-silico, ex-vivo, and in-vivo studies. *Journal of Drug Delivery Science and Technology*, 2023, 89, 105102.

17. Neha Sharma, Rasdeep Kour, Kommu Nagaiah, Shafiul Haque, Hardeep Singh Tuli, Sandeep Kaur and Satwinderjeet Kaur*. *Plumbago zeylanica L.* root fraction mitigates oxidative stress-induced DNA damage in human blood lymphocytes and exhibits cytotoxicity against MCF-7 cells. *Journal of Biological Regulators and Homeostatic Agents*, 2023, 37(10), 5489–5502.

18. Shubham Thakur, Amrinder Singh, Manjot Kaur, Navid Reza, Nitish Kumar, Rasdeep Kour, Satwinderjeet Kaur, Preet Mohinder Singh Bedi, Subheet Kumar Jain*. Vitamins and minerals fortified emulsion of omega-3 fatty acids for the management of preterm birth: In-vitro, in-silico, and in-vivo studies. *Journal of Drug Delivery Science and Technology*, 2023, 83, 104409.

19. Gera Narendra, Baddipadige Raju, Himanshu Verma, Manoj Kumar, Subheet Kumar Jain, Gurleen Kaur Tung, Shubham Thakur, Rasdeep Kour, Satwinderjeet Kaur, Bharti Sapra and Om Silakari*. Scaffold hopping based designing of selective ALDH1A1 inhibitors to overcome cyclophosphamide resistance: synthesis and biological evaluation. *RSC Medicinal Chemistry*, 2023, 15(1), 309–321.

20. Shubham Thakur, Rasdeep Kour, Satwinderjeet Kaur and Subheet Kumar Jain*. Spray-Dried Microspheres of Carboplatin: Technology to Develop Longer-Acting Injectable with Improved Physio-Chemical Stability, Toxicity, and Therapeutics. *AAPS PharmSciTech*, 2022, 23(5), 128.

21. Sukhvinder Dhiman, Rasdeep Kour, Gulshan Kumar, Satwinderjeet Kaur, Vijay Luxami, Prabhpreet Singh and Subodh Kumar*. 'Turn-On' monopodal and dipodal nanoprobes for serum albumins–a case of shift in selectivity towards BSA and a Z-to-U-like conformational change. *Materials Chemistry Frontiers*, 2022, 6(18), 2651–2660.

22. Navdeep Kaur, Rajdeep Kaur, Rasdeep Kour, Satwinderjeet Kaur and Prabhpreet Singh*. Pseudo-Crown ether II: Intracellular and solution-, SiNPs based test-kits for ppm level detection of H_2S gas. *Dyes and Pigments*, 2022, 202, 110280.

23. Sukhvinder Dhiman, Rasdeep Kour, Satwinderjeet Kaur, Prabhpreet Singh and Subodh Kumar*. Mitochondria targeted dual-fluorescent probe for bio-imaging viscosity and F^- with different fluorescence signals. *Bioorganic Chemistry*, 2022, 129, 106169.

24. Rajdeep Kaur, Rasdeep Kour, Satwinder Singh Marok, Satwinderjeet Kaur and Prabhpreet Singh*. AIE+ ESIPT Active Hydroxybenzothiazole for Intracellular Detection of Cu^{2+}: Anticancer and Anticounterfeiting Applications. *Molecules*, 2022, 27(22), 7678.

25. Amrinder Singh, Shubham Thakur, Amardeep Singh Virdi, Narpinder Singh, Rasdeep Kour, Satwinderjeet Kaur, Subheet Kumar Jain*. Novel methylcellulose based thermosensitive in situ nano liposomes of docetaxel for improved pharmacokinetics and pharmacodynamics with reduced toxicity. *Materials Today Communications*, 2022, 33, 104167.

Book Chapters

1. Rasdeep Kour, Sukhmanpreet Kaur Bal, Simrandeep Kaur, Neha Sharma, Sandeep Kaur, and Satwinderjeet Kaur*. "Gut Microbiota and Immune System: Interaction and Impacts." In Gut Microbiota.
   Jenny Stanford Publishing, 2025, 109-132.

2. Simrandeep Kaur, Mehak Taneja, Rasdeep Kour, and Satwinderjeet Kaur*. "Gut Microbiota and

Diet: Food Conversions and Metabolism." In Gut Microbiota.
Jenny Stanford Publishing, 2025, 9-32.

3. Sandeep Kaur*, Gulshan Bhagat, Rasdeep Kour, Navdeep Kour, Simrandeep Kaur, Simran Singh, Shagun Verma, Astha Bhatia, and Satwinderjeet Kaur. "Role of Gut Microbiota in Cancer Prevention: Mechanisms and Evidence." In Gut Microbiota.
Jenny Stanford Publishing, 2025, 77-107.

4. Kritika Pandit, Ajay Kumar, Sandeep Kaur, Neha Sharma, Rasdeep Kour, Renu Bhardwaj, and Satwinderjeet Kaur. Mechanistic Insight into the Chemotherapeutic Potential of Dietary Phytochemicals. Functional Foods for Health Maintenance: Understanding their Role in Cancer Prevention.
Bentham Science Publishers, 2023, (20) 70-89.

Area of Expertise

- Cancer Chemoprevention
- In-vitro and In-vivo Cancer Models
- Phytochemical Isolation
- Animal Cell Culturing
- Phytomedicine
- Nano-formulations
- In-vitro Toxicology

Skills

Laboratory Techniques

- Animal Cell culturing: Maintenance of cell lines, viability, cytotoxicity assays, isolation of nucleic acids and proteins, preparation of cell line samples for various studies
- Proficient in Colocalization in cell biology studies.
- Molecular techniques: Western blotting, RT-qPCR, Gel electrophoresis
- Animal handling: Tumor induction, dose administration, animal care.
- Proficient in natural product isolations from plants, Chromatographic techniques, Liposomes and Phytosomes synthesis.

Analytical Instruments

- Microscopes: Light, Confocal and Fluorescence Microscopy
- Flow cytometer
- Dynamic Light Scattering (DLS)
- UV-vis spectrophotometer
- Multi-mode microplate readers

- High-performance liquid chromatography.

Softwares

Linux, Microsoft Windows, Microsoft Office, Libre Office, Origin Pro, Prism-Graphpad, LaTeX, IBM SPSS, PAST, C6 BD accuri, Image J, Chem Draw, Casp, NIS Elements AR analysis and Shimadzu Lab solutions.

Teaching Experience

Assistant Professor, Khalsa College for Women, Amritsar 21/07/2025 – Till date

The Researcher taught Plant physiology to Third-year undergraduates, and Plant diversity to First-year undergraduates.

Assistant Professor, Khalsa College of Education, Amritsar 29/01/2025 – 16/06/2025

The Researcher taught cell biology and genetics to First-year undergraduates and Ecology to Third-year undergraduates.